

Project Initialization and Planning Phase

Date	11 July 2026
Team ID	
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to develop an intelligent Machine Learning system that predicts the Human Development Index (HDI) category (Very High, High, Medium, or Low) based on key socio-economic indicators, enabling faster and more accurate development analysis..
Scope	The project focuses on collecting HDI data from reliable sources, preprocessing the data, training Machine Learning models, evaluating prediction accuracy, and deploying the best-performing model through a user-friendly Flask web application for real-time HDI prediction..
Problem Statement	
Description	Analyzing Human Development Index data manually is time-consuming and requires processing multiple socio-economic indicators. Existing methods provide statistical reports but do not offer automated prediction, making development analysis slower and more complex.

Impact	Solving these challenges enables quicker and more accurate HDI prediction, supports evidence-based policymaking, simplifies country comparison, and assists researchers and organizations in making informed development decisions..
Proposed Solution	
Approach	The system applies Machine Learning classification algorithms to analyze key indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita to predict the HDI category. The trained model is integrated into a Flask-based web application that provides real-time predictions.
Key Features	Machine Learning-based HDI category prediction. User-friendly web interface for entering indicator values. Real-time prediction of HDI category. Data visualization using charts and graphs. Model comparison using multiple Machine Learning algorithms. Secure and efficient prediction system. Easy deployment using Flask and Render.



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computing Resources	Intel Core i5 or higher / AMD Ryzen 5, optional GPU
Memory	Memory	Memory

Storage	Storage	Minimum 10 GB free disk space (SSD recommended)
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Pickle
Development Environment	IDE	Jupyter Notebook, Visual Studio Code / PyCharm
Data		
Data	Source, size, format	UNDP / World Bank / Kaggle HDI Dataset, CSV format (approximately 150–200 country records)